

ASSESSMENT-3
CSA1707- ARTIFICIAL INTELLIGENCE

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Q1. Smart Home Automation System using Intelligent Agents

Problem Understanding

A smart home automation system is designed to manage **lighting, temperature, and security** based on user behavior and environmental conditions. The system receives real-time inputs such as motion detection, temperature changes, and time of day. It also learns user preferences over time to improve comfort, energy efficiency, and security.

Types of Intelligent Agents

1. Simple Reflex Agent

A Simple Reflex Agent makes decisions based only on the **current percept** without considering past information.

Example in Smart Home:

- Turns ON lights when motion is detected.
- Turns OFF lights when no motion is detected.
- Activates an alarm when an intruder is detected.

Advantages

- Fast response
- Easy to implement

Disadvantages

- Cannot learn user preferences.
- Cannot handle complex situations.

2. Model-Based Agent

A Model-Based Agent maintains an **internal model** of the environment and uses past information to make decisions.

Example in Smart Home:

- Remembers whether lights are already ON or OFF.
- Tracks room temperature throughout the day.
- Identifies occupied and unoccupied rooms.

Advantages

- Handles partially observable environments.
- Makes more accurate decisions.

Disadvantages

- Requires additional memory.
- More complex than a simple reflex agent.

3. Goal-Based Agent

A Goal-Based Agent selects actions that help achieve a specific goal.

Example in Smart Home:

- Maintain room temperature at **24°C**.
- Ensure all doors are locked before bedtime.
- Reduce electricity usage while maintaining comfort.

Advantages

- Makes intelligent decisions based on goals.
- Can compare different actions.

Disadvantages

- Requires planning.
- Decision-making may take more time.

4. Utility-Based Agent

A Utility-Based Agent chooses the action that provides the **highest overall benefit (utility)**.

Example in Smart Home:

- Balances comfort and electricity consumption.
- Chooses the most energy-efficient temperature setting.
- Adjusts lighting based on occupancy and daylight.

Advantages

- Optimizes multiple objectives.
- Provides better overall performance.

Disadvantages

- Utility function can be difficult to design.
- Requires more computation.

Application of Concepts

The smart home system applies AI concepts as follows:

- **Percepts:** Motion sensors, temperature sensors, security cameras, time of day.
- **Actions:** Turn lights ON/OFF, adjust temperature, lock/unlock doors, activate alarms.
- **Environment:** Partially observable, dynamic, sequential, and single-agent.
- **Learning:** Improves decisions by learning user preferences over time.

Best Agent / Hybrid Approach

A **Hybrid Agent (Model-Based + Goal-Based + Utility-Based)** is the most effective solution.

Reason

- The **Model-Based Agent** remembers the current state of the home.
- The **Goal-Based Agent** achieves objectives such as maintaining comfort and security.
- The **Utility-Based Agent** selects the best action by balancing comfort, energy efficiency, and safety.

Together, these agents provide intelligent, adaptive, and efficient home automation.

Justification

A hybrid approach is the most suitable because it combines memory, planning, and optimization. It can learn user behavior, respond to real-time events, reduce energy consumption, and maintain a secure and comfortable home environment.

Result

The Smart Home Automation System can effectively manage lighting, temperature, and security by using a **Hybrid Intelligent Agent** consisting of **Model-Based, Goal-Based, and Utility-Based Agents**. This approach ensures better comfort, energy efficiency, and security while adapting to user preferences over time.

Q2. Robot Navigation in an Unknown Maze using UCS and IDS

Problem Understanding

A robot is placed in an **unknown maze** and must find a path from the **Start (S)** to the **Goal (G)** without prior knowledge of the maze. The robot must explore the environment, avoid obstacles, and determine the best path using **Uninformed Search Strategies** such as **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)**.

Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the **lowest cumulative path cost** first. It guarantees the optimal solution when all path costs are non-negative.

Advantages

- Always finds the optimal path.
- Considers different path costs.
- Suitable for weighted graphs.

Disadvantages

- Requires more memory.
- Slower when many nodes have similar costs.

Time Complexity

- $O(b^{(1 + \lceil C/\epsilon \rceil)})^*$

Space Complexity

- $O(b^{(1 + \lceil C/\epsilon \rceil)})^*$

Where:

- **b** = branching factor
- **C*** = optimal path cost
- **ε** = minimum step cost

Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs Depth First Search with increasing depth limits until the goal is found.

Advantages

- Requires less memory.
- Complete for finite search spaces.
- Suitable when solution depth is unknown.

Disadvantages

- Repeats node expansions.
- Does not consider path costs.

Time Complexity

- $O(b^d)$

Space Complexity

- $O(b \times d)$

Where:

- **b** = branching factor
- **d** = depth of the solution

Comparison of UCS and IDS

Feature	UCS	IDS
Search Type	Cost-based	Depth-based
Optimal	Yes	Yes (equal costs)
Complete	Yes	Yes
Time Complexity	$O(b^{(1+\lceil C^*/\epsilon \rceil)})$	$O(b^d)$
Space Complexity	High	Low
Best Use	Weighted graphs	Unknown depth problems

Relation to Intelligent Agents

Simple Reflex Agent

- Makes decisions using only current observations.
- Cannot solve unknown mazes effectively.

Model-Based Agent

- Stores previously visited locations.
- Builds an internal map while exploring.

Goal-Based Agent

- Plans actions to reach the exit.
- Selects paths that lead toward the goal.

Utility-Based Agent

- Chooses the path with the maximum benefit.
- Considers safety, cost, and efficiency.

Effect of Agent Design on Decision Making

The type of intelligent agent determines how the robot makes decisions.

- A **Simple Reflex Agent** cannot remember previous states.
- A **Model-Based Agent** updates its knowledge as it explores.
- A **Goal-Based Agent** focuses on reaching the exit.
- A **Utility-Based Agent** selects the safest and least-cost path.

Thus, more advanced agents make better decisions in unknown environments.

Best Combination

Model-Based Goal-Based Agent + Uniform Cost Search (UCS)

Justification

- The **Model-Based Agent** stores information about explored paths.
- The **Goal-Based Agent** plans actions to reach the exit.
- **Uniform Cost Search** finds the minimum-cost path.

This combination enables the robot to explore efficiently, remember previous states, avoid unnecessary paths, and reach the goal with the lowest possible cost.

Result

The robot can successfully navigate an unknown maze using **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)**. Among these, **Uniform Cost Search combined with a Model-Based Goal-Based Agent** is the most suitable approach because it remembers explored paths, plans intelligently, and guarantees the least-cost path to the exit.